NGUYEN VAN HAI

BACKEND ENGINEERING

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LeetCode Profile
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About me

My name is Nguyen Van Hai. I graduated with a degree in Information Technology from Ho Chi Minh City University of Science - HCMUS, very passionate about technology and system architecture. I am a quick learner with strong problem-solving skills who thrive in a collaborative environment to achieve common goals. I have experience working on POS systems and LMS platforms using Golang and Node.js, along with DevOps practices to build scalable and high-performance application systems.

Education

•VNUHCM - University of Science *2020 - 2024*

Bachelor in Information Technology, Faculty of Information Technology – View degree GPA: 3.2 / 4

Skills

•Technical Skills

(Technologies, Cloud Infrastructure, and AI Systems)

- **Programming Languages**:
  - **Golang**: Experienced in building high-throughput microservices using Gin-Gonic, GORM, SQLX, gRPC, and concurrency patterns.
  - **JavaScript / TypeScript**: Experience with ExpressJS, NestJS, Prisma, Sequelize, gRPC, GraphQL, and frontend development with ReactJS, NextJS, and NextAuth.
  - **Python & Others**: Python (FastAPI, automation scripts), PHP (Moodle development, PHPUnit, Behat), and SQL.
- **Cloud Infrastructure & DevOps**:
  - **Cloud & IaC**: Azure (AKS, Event Hubs, Azure Redis Cache, Azure AI Foundry, Blob Storage, VNet/NSG), Terraform (Infrastructure as Code).
  - **Container Orchestration & GitOps**: Kubernetes (AKS, k0s), Helm, Docker, ArgoCD (GitOps App-of-Apps, automated pre-upgrade DB migration hooks, Cronjob).
  - **Networking & Edge Security**: NGINX Ingress Controller, Cloudflare (DNS, WAF, Edge SSL).
  - **Observability & Performance**: Grafana, Datadog, Langfuse (LLM tracing), and stress testing with k6.
- **Databases & Messaging**:
  - **Databases**: PostgreSQL, MongoDB, Redis, and schema migrations using Goose.
  - **Messaging & Events**: Azure Event Hubs / Apache Kafka (Kafka API protocol, Transactional Outbox pattern, partitioned event streaming).
- **AI Integration & Agentic Workflows**:
  - **Agentic SDLC & Multi-Agent Workflows**: Architected multi-agent pipelines using specialized subagents across the complete software development lifecycle (architecture/infra discovery, requirement clarification, implementation, automated testing, lint/security checks) leveraging Google Antigravity and OpenCode; codified personal engineering standards and guidelines into living AI Wiki/rulesets.
  - **Azure AI Foundry & LLMs**: Integrated Mistral OCR and GPT-4o vision for receipt intelligence; designed multi-agent automated reporting systems; integrated Ollama gateway for internal AI features.
  - **Vector Search & RAG**: Embeddings (text-embedding-3-small), QdrantDB, CockroachDB vector, and PostgreSQL pgvector for semantic search and abbreviation resolution.
- **Algorithms and Data Structures**: Strong foundations in algorithms, system design, and distributed data structures for high-concurrency systems.

Work experiences

•Backend & DevOps Engineer at Solum *02/2026 – Present*

Smart Retail POS Platform (Multi-Tenant Omnichannel POS & ERP)

- **Backend Engineering & Integrations**: Developed core payment, inventory, and order processing microservices in Golang; integrated Viettel & MISA E-Invoice APIs for automated per-order tax filing; implemented B2B ordering workflows, pay-later schemes, and shipment processing.

- **DevOps, GitOps & Cloud Infrastructure**: Codified Azure infrastructure using Terraform (IaC) and orchestrated AKS workloads via Helm; built GitOps CI/CD pipelines with ArgoCD (App-of-Apps pattern) featuring automated pre-upgrade DB migration hooks; automated wildcard SSL/TLS certificate issuance using cert-manager with Cloudflare APIs.
- **Event-Driven Architecture & Data Reliability**: Designed decoupled asynchronous messaging using Azure Event Hubs (Kafka protocol) with the Transactional Outbox pattern; managed PostgreSQL schema migrations via Goose and implemented Redis for distributed locking and session caching.
- **Performance, Networking & Observability**: Configured NGINX Ingress and Cloudflare for load balancing, WAF, and DNS routing; executed stress testing with k6 to eliminate bottlenecks by checking Grafana, kubectl and Datadog, ensuring seamless zero-downtime rollouts across **200+ retail stores**. Automated daily order reconciliation alerts - via slack bot webhook.
- **AI & Multi-Agent Systems**: Integrated receipt and document OCR using Mistral OCR on Azure AI Foundry, architected an AI-powered reporting solution using a multi-agent framework (data retrieval and analysis agents) powered by Azure GPT-4o.
- **Technical Stack**: Golang, PostgreSQL, Redis, Apache Kafka / Azure Event Hubs, Docker, Kubernetes (AKS, k0s), Helm, Terraform.
- **Tools & Cloud**: Azure (AKS, Event Hubs, Redis Cache, AI Foundry), ArgoCD, Cloudflare, NGINX Ingress, cert-manager, Grafana, Datadog, k6, Langfuse.
- **Role**: Backend / DevOps Engineer

•Javascript, Typescript and PHP Developer At Nashtech *3/2024 - 2/2026*

(Graduated from the Rookie to Engineering program - Batch 7 and transitioned to working full-time afterward)

- **Open University Moodle**: A Learning Management System (LMS) for university students and lecturers. Lecturers can create courses, quizzes, and assessments, while students can submit assignments and participate in class forums. The system is built on the LAMP stack and customized from Moodle open-source to meet Open University (OU) requirements. Developed and maintained Moodle plugins using PHP and JavaScript (e.g., OUStudio, Sketch, Assignment). Implemented unit tests (PHPUnit) and Behat tests for quality assurance. Integrated AI-powered Moodle plugins using Ollama AI to summary quiz and exam to user/teacher, and improved performance and reduced token usage by implementing Redis caching.
- **Technical Stack**: PHP, Moodle, PHPUnit, Behat test, Ollama AI
- **Internal Tools – NashTech**: Rookie to Engineer Project – Asset Management, Worked in a team as a Backend, DevOps Engineer. Responsible for designing and implementing CI/CD workflows, managing cloud infrastructure using Terraform (Infrastructure as Code), and developing backend services with NestJS. Worked with Azure services including Azure Web Apps, Azure Database for PostgreSQL, and Azure Pipelines to support deployment and infrastructure automation.
- **Technical Stack**: Back End (NestJS, Prisma) and Front End with ReactJS
- **Tools**: Azure DevOps, PostgreSQL, Azure Container Registry, Azure App Service
- **Role**: Fullstack Developer

Languages

–**Vietnamese:** Native proficiency.

–**English:** Fluent reading and writing; basic spoken communication.

Achievements

–**Gemini Certified Educator:** View Certificate

–**GitHub Certificate:** View Certificate

–**ASP.NET Core Developer:** View Certificate

–**Azure AI Essentials:** View Certificate